

Holographic Image Compression via Sparse Representation and Orthogonal Pursuit in Transformed Spaces

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Abstract—This paper introduces an innovative image compression framework based on Compressed Sensing theory and sparse representation. The proposed algorithm utilizes an iterative decomposition via Orthogonal Matching Pursuit (OMP) applied to an overcomplete dictionary generated through Kronecker products of DCT bases. The system is highly configurable, allowing for a dynamic trade-off between fidelity and size. Compared to standard formats, the method achieves a baseline compression ratio of 28:1 with a PSNR of 35.85 dB, while maintaining the capability to scale up to 150:1 for ultra-low-bandwidth scenarios.

I. INTRODUCTION

Classical compression methods based on the Discrete Cosine Transform (DCT) often suffer from blocking artifacts at high compression ratios. The *Sparsity* paradigm suggests that a signal can be faithfully reconstructed from a limited number of coefficients if projected onto an appropriate dictionary. This paper describes the implementation of a “holographic” encoder that treats the image as a superposition of frequency planes extracted iteratively through greedy optimization, providing a scalable solution for varying transmission requirements.

II. MATHEMATICAL FOUNDATIONS

A. Dictionary Construction $\mathbf{D}$

Let $\mathbf{D}_{1D} \in \mathbb{R}^{n \times n}$ be the Type-II DCT transform matrix. To extend the representation to the two-dimensional space of an image patch $P \times P$, we define the holographic dictionary using the Kronecker product:

$$\mathbf{D}_{2D} = \mathbf{D}_{1D} \otimes \mathbf{D}_{1D} \quad (1)$$

The final dictionary Φ is a block-diagonal matrix that operates across luminance and chrominance channels (Y, Cb, Cr), significantly reducing inter-channel redundancy.

B. Orthogonal Matching Pursuit (OMP) Algorithm

The encoding problem consists of finding a coefficient vector $\mathbf{x}$ such that:

$$\min \|\mathbf{y} - \Phi \mathbf{x}\|_2^2 \quad \text{subject to} \quad \|\mathbf{x}\|_0 \leq K \quad (2)$$

where $\mathbf{y}$ is the linearized patch vector and K is the number of desired “holographic planes.” The iterative selection of atoms ensures that the most energetic components of the signal are preserved first.

III. IMPLEMENTATION AND CONFIGURABILITY

A. Parameter Tuning and Scalability

A core strength of the proposed architecture is its extreme modularity. Users can fine-tune the compression behavior by adjusting three primary parameters:

- **K-Planes (K):** Controls the sparsity of the representation.
- **Patch Size (P):** Defines the local correlation window.
- **Quantization Bits (b):** Determines the precision of the stored coefficients.

B. Two-Pass Global Quantization

The algorithm implements a post-OMP global quantization strategy. The maximum absolute value of all coefficients across the entire page (α_{max}) is extracted, and each coefficient x is mapped to a b -bit integer:

$$q = \text{round} \left(\frac{x \cdot (2^{b-1} - 1)}{\alpha_{max}} \right) \quad (3)$$

IV. EXPERIMENTAL RESULTS AND DISCUSSION

Experimental tests conducted on high-resolution RAW scans demonstrate a highly flexible Pareto front.

A. Performance Trade-offs

The system allows for two extreme modes of operation:

- 1) **High-Fidelity Mode:** By increasing K and b , the algorithm achieves near-lossless reconstruction with superior PSNR, albeit with lower reduction ratios.
- 2) **Extreme Compression Mode:** By aggressive tuning of the sparsity constraints, the system can achieve **compression factors exceeding 150x**. While this results in a lower PSNR, the holographic nature of the reconstruction prevents the typical “blockiness” of JPEG, preserving the structural legibility of the document.

B. Baseline Benchmark

For a standard configuration (8×8 patches, $K = 12$), the system achieves a **28x compression ratio** with a **PSNR of 35.85 dB**, outperforming traditional encoders in high-frequency detail retention (e.g., text edges).

V. CONCLUSION

Daniele Rufo's algorithm proves that combining OMP with Kronecker-based dictionaries offers a superior path for extreme image and document compression. The ability to shift from 28x to over 150x compression without structural failure makes this system a robust candidate for the next generation of communication protocols and long-term digital archiving.